



PROBABILITY AND STATISTICS

QUESTION BANK

MODULE 1

SHORT QUESTION AND ANSWER 2 MARKS

1. Calculate mean from the following data

x	5	10	15	20
f	4	6	8	2

2. Define mean.

3. Write a short note on Geometric Mean .

4. Write a short note on Harmonic Mean.

5. Define Range and Coefficient of Range.

6. Find out the median of the following items. X: 10, 15, 9, 25, 19.

7. Define median

8. Define Mode. Find the mode of 1, 2, 3, 4, 1, 1, 1, 5, 2, 2, 3, 1, 1

9. Find the IQR for the data: 2, 4, 6, 8, 10, 12, 14, 16, 18

10. Find the range of weights of 7 students from the following. 27, 30, 35, 36, 38, 40, 43

LONG QUESTION AND ANSWERS (6 MARKS)

1. What is Statistics? Characteristics of Statistics, merits and demerits.

2. From the following find out the mean using assumed mean method:

Class Interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	6	5	8	15	7

3. Calculate the arithmetic mean using the Shortcut Method from the following data:

Marks (x)	2	4	6	8	10	12	14
No. of Students (f)	11	9	7	10	6	4	3

4. Consider the following distribution of daily wages of 50 workers of a factory.

Daily wages (in ₹)	500-520	520-540	540-560	560-580	580-600
NO. OF WORKERS	12	14	8	6	10

Find the mean daily wages of the workers of the factory by using an appropriate method.



5. Find the median for the following data

x	2	4	6	8	10	12	14
f	4	6	10	12	8	7	3

6. From the following find out the mode

Class Interval	10-25	25-40	40-55	55-70	70-85	85-100
Frequency	6	20	44	26	3	1

7. Find out the geometric mean of the following data.

Yield of wheat (mounds)	No. of farms
7.5 – 10.5	5
10.5 – 13.5	9
13.5 – 16.5	19
16.5 – 19.5	23
19.5 – 22.5	7
22.5 – 25.5	4
25.5 – 28.5	1

8: Find IQR for the following data.

Class Intervals	0-10	10-20	20-30	30-40	40-50
Frequencies	5	8	12	10	5

9. Compute the standard deviation from the following data:

Class	0 - 10	10 - 20	20 - 30	30 - 40	40-50
Frequency	5	8	15	16	6



10. Calculate the standard deviation AND variance from the following data;

Marks	10	20	30	40	50	60
No. of students	8	12	20	10	7	3

MODULE 2

SHORT QUESTION AND ANSWER 2 MARKS

1. What is Correlation?
2. State the Range of Correlation Coefficient.
3. What is Regression?
4. Write the formula for Karl Pearson's correlation coefficient.
5. What are Regression Lines ? Define them.
6. If $b_{xy} = 0.25$ and $b_{yx} = 0.64$, find r .
7. Define Rank correlation
8. Define Relation between Correlation and regression.
9. Write the advantages of Regression.
10. Write the formula of Spearman's Rank Correlation

LONG QUESTION AND ANSWERS (6MARKS)

1. Define correlation and explain different types of correlation.
2. Find Karl Pearson's coefficient of correlation from the following data

Wages	100	101	102	102	100	99	97	98	96	95
Cost of living	98	99	99	97	95	92	95	94	90	91

3. Following are the rank obtained by 10 students in two subjects, Statistics and Mathematics. To what extent the knowledge of the students in two subjects is related?

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	2	4	1	5	3	9	7	10	6	8

4. Determine the equation of a straight line which best fits the data.



X:	10	12	13	16	17	20	25
Y:	10	22	24	27	29	33	37

5. Calculate the two regression equations of y on x and x on y

X	1	2	3	4	5
Y	2	3	5	4	6

6. Calculate the regression equations of Y on X from the data given below, taking deviations from actual means of X and Y

Price (RS.)	10	12	13	12	16	15
Amount Demanded	40	38	43	45	37	43

7. Explain the differences between correlation and Regression.

8. Calculate Spearman's rank correlation coefficient for the data

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

9. From a sample of 200 pairs of observations, the following quantities were calculated.

$\Sigma X = 11.34$, $\Sigma Y = 20.78$, $\Sigma X^2 = 12.16$, $\Sigma Y^2 = 84.96$, $\Sigma XY = 22.13$ From the above data show how to compute the coefficients of the equation $Y = a + bX$.

10: Fit a second degree polynomial to the following data by the method of least squares:

x	10	12	15	23	20
y	14	17	23	25	21



MODULE 3

SHORT QUESTION AND ANSWER 2 MARKS

1. Define Experiment and Random Experiment.
2. Define Probability.
3. State the Addition Theorem of Probability.
4. In a class there are 10 boys and 5 girls. A committee of 4 students is selected at random. Find the probability that the committee contains at least 3 girls.
5. Define Conditional Probability. Give an example.
6. State the Multiplication Theorem of Probability.
7. What is the probability that a leap year has 53 Sundays?
8. If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, and $P(A \cup B) = \frac{1}{2}$, find:
 - (i) $P(B/A)$
 - (ii) $P(A/B)$
9. Determine the probability that a non-defective bolt will be found if, out of 600 bolts already examined, 12 were defective.
10. Two dice are thrown. Let A be the event that the sum of the points is 9 and B be the event that at least one number is 6. Find:
 - (i) $P(A \cap B)$
 - (ii) $P(A \cup B)$
 - (iii) $P(A' \cup B')$

LONG QUESTION AND ANSWERS (6MARKS)

1. A class consists of 6 girls and 10 boys. If a committee of 3 members is chosen at random, find the probability that:

- (i) 3 boys are selected
- (ii) Exactly 2 girls are selected.



2.Explain the Axiomatic Definition of Probability and state its axioms.

3.Two cards are selected at random from 10 cards numbered 1 to 10. Find the probability that the sum is odd when:

- (i) Two cards are drawn together.
- (ii) Two cards are drawn one after another with replacement.
- (iii) Two cards are drawn one after another without replacement.

4.If $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, and $P(A \cap B) = \frac{1}{5}$, find:

- (i) $P(A \cup B)$
- (ii) $P(A' \cap B)$
- (iii) $P(A \cap B')$
- (iv) $P(A' \cap B')$

5.Box A contains 5 red and 3 white marbles and Box B contains 2 red and 6 white marbles. If a marble is drawn from each box, find the probability that both marbles are of the same colour.

6.From a city, 3 newspapers A, B and C are published. A is read by 20%, B by 16%, C by 14%, both A and B by 8%, both A and C by 5%, both B and C by 4%, and all A, B and C by 2%. Find the percentage of the population that reads at least one paper.

7.In a group there are 3 men and 2 women. Three persons are selected at random. Find the probability that:

- (i) One man and two women are selected.
- (ii) Two men and one woman are selected.

8.Three machines 1, 2 and 3 produce 40%, 30% and 30% respectively of the total number of items in a factory. The percentages of defective items produced by these machines are 4%, 2% and 3% respectively. If an item is selected at random, find the probability that the item is defective.

9. A bag contains 4 red and 5 black balls, Find the probability of drawing 2 red balls in succession when the first ball drawn is:

- (i) Not replaced.
- (ii) replaced.

10.A class has 20 boys and 5 girls. Three students are selected at random one after another. Find the probability that:

- (i) The first two are boys and the third is a girl.
- (ii) The 1st and 3rd are of the same gender and the 2nd is of the opposite gender, without replacement.



MODULE 4

SHORT QUESTION AND ANSWER 2 MARKS

1. A fair coin is tossed six times. Find the probability of getting four heads.
2. State the mean and variance of a binomial distribution.
3. Write the probability mass function of a Poisson distribution.
4. State any two properties of a probability distribution.
5. The probability mass function of the random variable X is given below:

X	0	1	2	3	4	5	6
$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$

Find k ?

6. A continuous random variable X has the probability density function $f_X(x) = \begin{cases} \frac{1}{6}, & -3 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$. Find

$P(X < 1)$.

7. A Random variable is uniformly distributed over the interval 2 to 10. Find its Variance.
8. Define probability density function.
9. If the probability of a defective bolt is 0.2, find (i) mean (ii) standard deviation (S.D.) for the distribution of bolts in a total of 400.
10. A die is thrown thrice. A success is getting 1 or 6 on a thrown. Find the mean of the number of successes.

LONG QUESTION AND ANSWERS (6MARKS)

1. Define the binomial distribution and derive the mean.
2. Determine the binomial distribution for which the mean is 4 and variance is 3.
3. A manufacturer of pins knows that 2% of his product is defective. If he sells pins in boxes of 100 and guarantees that not more than 4 pins will be defective, what is the probability that a box will fail to meet the guaranteed quality?
4. If X is a uniformly distributed random variable with mean 1 and variance $\frac{4}{3}$, find $P(X < 0)$.
5. Derive the Mean of the Continuous Uniform distribution.
6. Define the properties of probability density function and distribution function.
7. A continuous random variable X has the probability density function $f_X(x) = \begin{cases} Kx, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$. Find:
 - i. The value of the constant K so that $f_X(x)$ is a valid probability density function.
 - ii. The corresponding Cumulative Distribution Function (CDF).



8. The probability distribution of a discrete random variable X is given below:

X	0	1	2	3	4	5	6	7
$P(X)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Find:

- The value of K .
- $P(X \leq 3)$.
- The minimum value of m such that $P(X \leq m) > 0.5$.

9. Find the Standard Deviation of a Continuous Uniform Distribution over the interval $[0,1]$.

10. Three light bulbs are chosen at random from 12 bulbs of which 5 are defective. Find the probability that:

- All are defective
- One is defective

MODULE 5

SHORT QUESTION AND ANSWER 2 MARKS

- Definition of Population and Sample?
- Define Parameter and Statistics?
- Find the value of the finite population correction factor for $n = 10$ and $N = 1000$.
- The variance of the population is 2, if the size of the sample collected from their population is 169. find the standard error of sample mean.
- Define Central Limit Theorem.
- If $\alpha = 0.01$, what is the probability of making a Type I error?
- A test gives P-value of 0.03 at a significance level of 0.05. Should H_0 be rejected?
- A sample mean is 72, $\sigma = 10$, $n = 100$ and $z = 1.96$. Find the 95% confidence interval.
- A population has $\mu = 60$, $\sigma = 12$ and $n = 36$. Find the probability that the sample mean lies between 58 and 62.
- A population has $\mu = 80$ and $\sigma = 15$. For $n = 25$, find $E[X]$.

LONG QUESTION AND ANSWERS (6MARKS)

- A population consists of 5 numbers: 2, 3, 6, 8, 11

Find:

- Mean of the population
- Standard deviation of the population

- A population consists of 6 numbers: 4, 8, 12, 16, 20, 24. Consider all possible samples of size '2' that can be drawn without replacement from this population. Find:

- Mean of sampling distribution of means.
- S.D. of sampling distribution of means.



3. The mean height of the students in a college is 155 cms and standard deviation is 15. What is the probability that mean height of 36 students is less than 157 cms?
4. A random sample of size 100 is taken from an infinite population having the mean $\mu = 76$ and $\sigma^2 = 256$. What is the probability that $\bar{X}$ will be between 75 and 78?
5. Write the types of hypotheses?
6. A machine is designed to produce insulating washers for electrical devices of average thickness of 0.025 cm. A random sample of 10 washers was found to have a thickness of 0.024 cm with a S.D. of 0.002 cm. Test whether the sample mean differs significantly from the population mean at the 5% level of significance. The critical value of t for 9 degrees of freedom at 5% level is 2.262.”.
7. A sample of 26 bulbs gives mean life of 990 hours with S.D. of 20 hours. The manufacturer claims that the mean life of bulbs is 1000 hours. Is the sample up to the standard?
8. Write the procedure about t-test for difference of means.
9. Write the procedure about F-test for small samples.
10. Pumpkins were grown under 2 experimental conditions. Random samples of 11 and 9 pumpkins show the S.D. of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test hypotheses that the 2 variances are equal.